VIGNESH S R

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OBJECTIVE

Third-year B.Tech ECE student with hands-on experience in embedded systems, PCB design, and Verilog, seeking a core internship in embedded, hardware, or semiconductor engineering.

EDUCATION

B.Tech in Electronics and Communication Engineering , Vellore Institute of Technology 2024 - 2028

CGPA : 9.20/10

SKILLS

Embedded	Embedded C, Arduino, ESP32/ESP8266, Timers, Interrupts, UART, SPI, I2C
Hardware	PCB Design, Circuit Design, Sensor Interfacing, Hardware Debugging, Soldering
Digital/VLSI	Verilog HDL, ModelSim, Quartus Prime
Tools	KiCad, Keil uVision, Multisim, MATLAB
Comm/IoT	GNSS/GPS, NavIC, GIS, MQTT, IoT
Programming	C, Python, Java

INTERNSHIPS

Hardware Intern — Elena Geo Tech Private Limited, Bengaluru Jun 2026
KiCad PCB design, soldering, board-level debugging, and GNSS/NavIC integration with embedded and IoT systems.

Embedded Systems Intern — Maven Silicon May 2026
Embedded C, microcontroller programming, UART/SPI/I2C, timers, interrupts, and hardware-software integration.

PROJECTS

SAKTHI – Smart Agricultural Monitoring Robot

NavIC/GIS-enabled autonomous rover with AI-based disease detection, irrigation advisory, and real-time MQTT dashboard.

SOS Emergency Alert Device

Embedded GPS/GSM device transmitting real-time location and emergency alerts to predefined contacts.

Multipurpose Agri Rover (MARCO)

IoT rover with DHT11/soil moisture sensors, seed dispenser, and ESP32-CAM for crop health analysis with wireless control.

Sign Language Capture & Illustration System

Smart glove with flex sensors, IMU, and ML to translate gestures into real-time text and visuals.

Dual-axis Solar Tracker

Servo-controlled sun-tracking system improving energy harvesting over fixed installations.

Performance Analysis & Handover in Hybrid Wi-Fi/Li-Fi Systems

Simulated a hybrid network with seamless handover, improving connectivity and reducing latency.

CERTIFICATIONS

MATLAB Onramp (Oct 2025) | Signal Processing Onramp (Oct 2025) | Nokia Campus Connect 2025 (Jul 2025)